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Prioritizing the key causes of construction project delay in different countries: a cross-sectional analysis of different project types

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ABSTRACT

The construction industry is a key sector driving global economic growth. Despite advancements in managerial models, it continuesto face challenges, notably delays and underperformance in projects, a persistent issue worldwide. This study offers a comprehensive review and analysis of primary causes of delays in construction projects across various countries, using the PRISMA method. The research makes key contributions: a) Identifying primary causes of delays in different project types, including residential buildings, public buildings, transport infrastructure, and other infrastructure projects; b) Comparing delay causes across 35 countries; c) Evaluating the most commonly used decision-making models for prioritizing delay risk factors; and d) Ranking causes of delays in construction projects based on 92 studies. Findings show that while delay causes vary by country and project type, certain factors—such as financial issues with the owner and delayed contractor payments—are widespread. Transport infrastructure projects are particularly affected by poor site conditions, topographical challenges, and weak project management. Residential projects often face delays due to changes in owner requirements and execution errors, while public buildings and other infrastructure projects suffer from inaccurate time estimations. This study provides valuable insights into the complexities of project delays and offers guidance for improving project management practices to mitigate delays.

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1. Introduction

On a global scale, the construction industry is considered essential, as it serves as the cornerstone of development for all countries and provides a platform for the emergence of various other industries (Mahamid 2012). While the construction industry is the runner-up in terms of economic significance, right after agriculture, investors do not find it an attractive sector financially, due to the low level of return on investment (Aboseif and Hanna 2023). The construction industry is one of the primary sectors whose impact on economic growth is substantial and cannot be underestimated. Construction serves as a means through which a society achieves its goals in terms of urban and rural development. This achievement is attributed to the complex network of communications it has established with other industries (Wali and Ali 2019).

In recent years, even with advancements in managerial models, construction projects continue to encounter significant challenges in meeting performance benchmarks. Notably, project delays—integral to assessing performance—remain a pervasive issue, undermining project success. There are still a lot of obstacles that frequently obstruct the efficient advancement of building projects (Patasik et al. 2022). Despite extensive scientific research and improved management techniques, the construction industry still grapples with persistent issues such as delays, conflicts, productivity losses, subpar performance, coordination challenges, and more. The problem of underperformance in construction projects is a

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global issue (Bertelsen and Sacks 2007). Prominent projects are sometimes in the news for all the wrong reasons: confusing cost overruns, protracted delays, and a plethora of underwhelming benefit performances (Ika et al. 2023). Regardless of sector or location, project failures are predicted to cost billions of Euros yearly (Joslin and Müller 2016). Still, over 30% of projects fail to satisfy quality requirements, run over budget, or miss deadlines (Bushuyev and Wagner 2014). Poor project execution results in the loss of over 12% of all investments (Institute 2016). Based on a survey in 2015, 77% of the mega projects with values over $200 M experienced more than 40% delay, and 30% of the projects spent more than planned budgets (Changali et al. 2015).

Schedule delays are a prevalent issue afflicting construction projects success (Alshibani et al. 2022). The success of projects is closely tied to their timelines, and any delays can significantly impact their overall success. Therefore, the timeframe of projects is a critical factor that cannot be overlooked (Zaman et al. 2022). Construction delays are a worldwide occurrence that impact not just the building sector but also the economies of entire nations (Sambasivan and Soon 2007). There are direct negative effects of building project delays on the economy and the standard of living of the populace. Construction projects are performing poorly as a result of delays and disagreements. Almost all construction projects experience delays, which can have a major impact on the project's success. Finding the true reasons for these delays is crucial to reducing or preventing them (Bekr 2015).

Considering the costs incurred due to project delays, this research aims to address the following key question: What are the primary causes of delay for each project type across different countries, and how do these causes vary between countries?

Numerous research studies and academic review papers have explored the factors contributing to construction project delays, with a focus on both international and developing nations (Bekr 2015; Alrasheed et al. 2023). Table 1 provides a comparative summary of previous meta-analyses on construction delays. While delay causes vary across countries and project types, only three prior studies (Rivera et al. 2020; Sanni-Anibire et al. 2022; Selcuk et al. 2024) explicitly compared results across different countries. Moreover, previous reviews either concentrated on a single project type or treated multiple types as a mixed category. To date, no study has systematically compared delay causes across distinct project types, leaving this as a clear gap in the literature. In addition, because prior reviews were based on a limited number of articles (ranging from 13 to 70), they evaluated the importance of at most 52 delay causes. By contrast, this research draws on a broader evidence base of 92 articles, identifies and analyzes 103 distinct delay factors, and extends the analysis across 35 countries. Uniquely, it provides a dual comparative perspective by examining delay causes both across countries and across project types. So, this study provides a comprehensive evaluation of the literature and by examining delay causes across different countries and project types, this study enables a comparative understanding of how delay factors vary by region and project category.

This research employs the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) method to conduct a comprehensive systematic review of the factors contributing to construction project delays (Moher et al. 2009). By systematically synthesizing data from a wide array of studies, this approach ensures a rigorous, transparent evaluation of the literature, enabling the identification and prioritization of delay factors across various project types (Moher et al. 2009).

This methodological rigour helps in pinpointing actionable strategies that can significantly improve efficiency in the construction sector. The process is divided into distinct phases: data collection and analysis, followed by evaluation and result presentation.

The data collection and analysis phase includes a meta-analysis to identify relevant studies related to the field and data extraction. The next phase is quantitatively analysing the data using BibExcel and R and performing statistical studies. Figure 1 presents the flowchart of the methodology adopted.

2. Materials and methods

The method includes two phases of data compilation and data analysis as below.

Table 1. Comparing the review papers.

Article	Year	number of articles	Number of factors	Project type					Multi-country analysis	Number of Countries	Compare the delay causes over	
				Residential	Public	Transport Infrastructure	Other Infrastructure	Not specify			Countries	Project Type
(Alrasheed et al. 2023)	2023	13	27	✓	✓				✗	1	✗	✗
(Selcuk et al. 2024)	2022	70	30	✓	✓				✓	33	✓	✗
(Sanni-Anibire et al. 2022)	2022	–	36					✓	✓	–	✓	✗
(Rivera et al. 2020)	2020	25	–			✓			✓	25	✓	✗
(Derakhshanfar et al. 2019)	2019	46	26					✓	✗	–	✗	✗
(Yap et al. 2021)	2019	26	52					✓	✗	1	✗	✗
This study	2025	92	103	✓	✓	✓	✓		✓	35	✓	✓

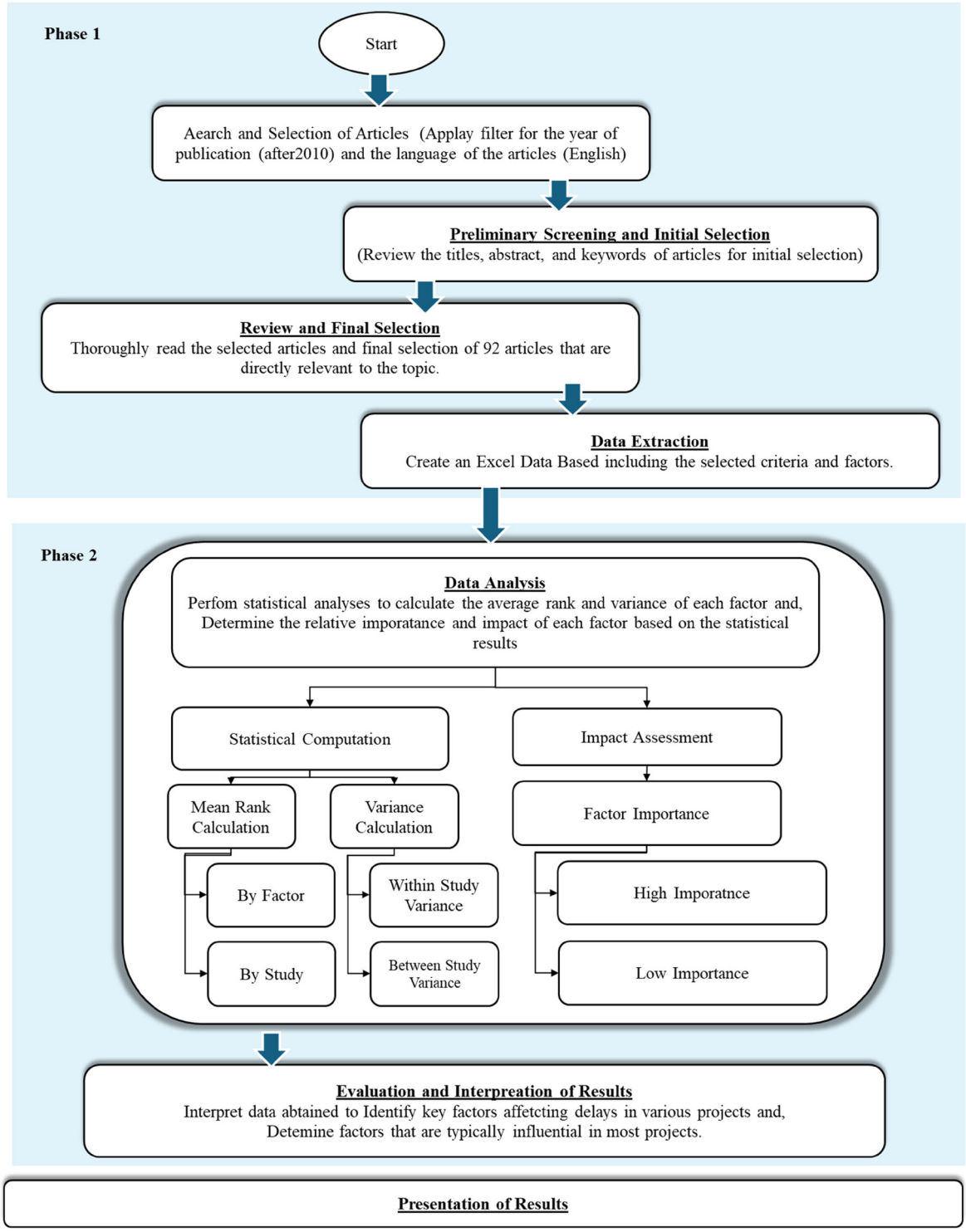


Figure 1. Flow chart of the methodology adopted for the review paper.

2.1. Phase 1: Data compilation and articles selection

This phase includes three steps of data collection and selection, macro perspective, and list of factors.

2.1.1. Data collection and selection of articles

Numerous models are presented in the literature to determine and prioritize the factors causing the project cost overrun. In this research, a comprehensive review is established on the key delay causes on



Figure 2. Number of publications on prioritizing project delay causes.

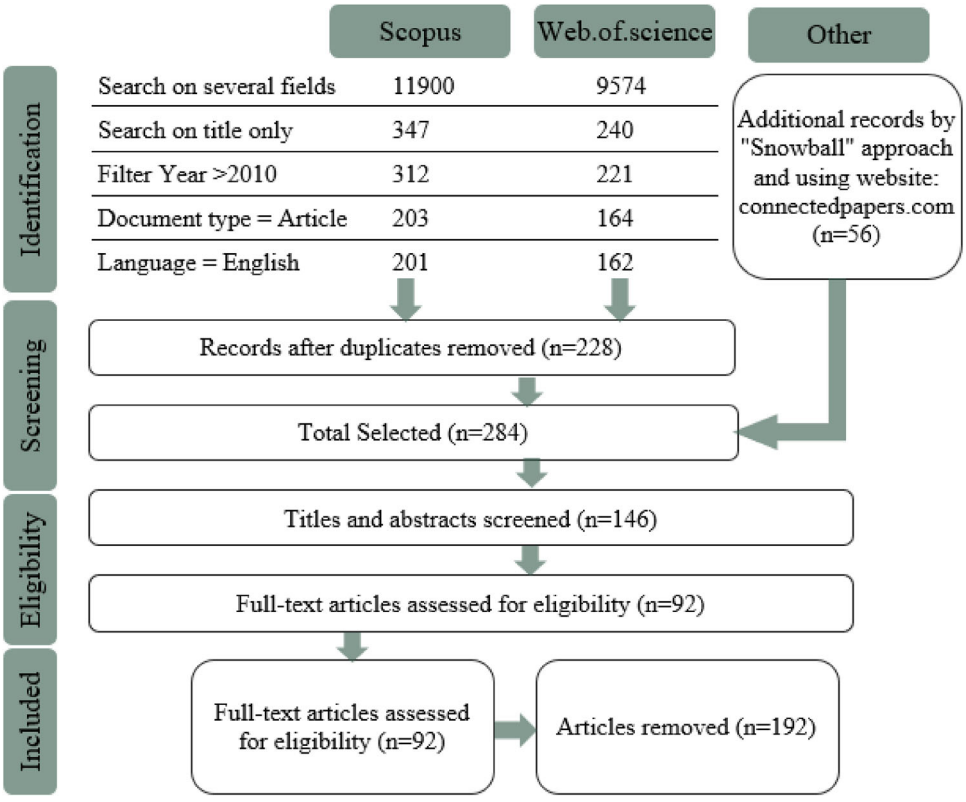


Figure 3. PRISMA model.

project management based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) method as presented in Figure 3 (Moher et al. 2009). This method includes 4 parts: identification, screening, eligibility, and inclusion.

In the first step, to capture the relevant studies, we adopted specific search strings in two major databases of Scopus and Web of Science. These two databases were selected because they are among the most comprehensive and widely recognized sources of peer-reviewed literature in the field of construction management. Their extensive indexing, rigorous selection criteria, and broad coverage ensure that a wide range of high-quality studies are included.

The exact search string adopted in databases was: '(project* OR construction) AND (delay OR 'Time Overrun') AND (factor* OR cause* rank* OR prioritiz* OR scor*)', which results a numerous articles. Figure 2 presents the increasing trend of the number of articles in this context, based on a structured search conducted in Scopus.

Searching with the specified keywords in titles, abstracts, and keywords yielded 11,900 articles in Scopus and 9574 articles in Web of Science. Further refinement was made by searching titles only, which narrowed the records to 347 in Scopus and 240 in Web of Science. Filtering was applied based on the year of publication (>2010), resulting in 312 records from Scopus and 221 from Web of Science. The rationale for excluding studies published before 2010 was to focus on more recent



Figure 4. The result of bibliometric analysis is based on the titles of the articles.

advancements in project delay analysis, as earlier studies might not reflect current industry practices, methodologies, and challenges.

The document type was then restricted to articles and the language filter was set to English, leaving 201 articles from Scopus and 162 from Web of Science. After removing duplicate articles, the total number of unique records reached 228. An additional 56 articles were also selected based on the snowball approach by using ConnectedPapers.com. Titles and abstracts were screened, further narrowing the pool to 146 records. Among all the 146 articles, 92 are selected after a full reading of articles.

2.1.2. Macro perspective

To see this research area from a macro perspective, and identify the most frequently used models, and studied country, bibliometric analysis was performed based on the article titles in the commonly published papers. The result is shown by the density maps in Figure 4. Based on the results, pipeline, gas, highway, and road are the most evaluated project types, and Iran, China, Iraq, Malaysia, Libya, and India are the main surveyed countries.

The word connection network visualization, representing the relationships between key terms in studies, is presented in Figure 5. The most prominent words include 'project management', 'delay time', 'construction industry', and 'surveys', suggesting that studies in this dataset focus on project delays, their causes, and management in the construction sector. In this figure, blue cluster includes terms like 'delay time', 'construction management', 'causes of delays', and 'importance index', indicating research on delay factors and their impact. Red cluster includes terms like 'surveys', 'questionnaire survey', and 'literature reviews' suggest methodologies used in these studies. Green and purple clusters include terms like 'costs', 'community development', and 'human resource management' show related research areas.

Determining the project delay causes is presented in several articles in the literature by using different decision-making models like the Analytic Hierarchy Process (AHP), Relative Importance Index (RII), and System Dynamic (SD). In this research, 92 refereed journal articles are selected and compared in terms of model, case study, country, and results are presented in Table A-1 in the supplementary material.

The distribution of countries in the selected studies is presented in Figure 6. Saudi Arabia, Malaysia, India, and China with respectively 8, 7, 6, and 4 studies are the most evaluated cases.

In this research, the most current articles in the ranking of project delay, based on the project types categorized into four groups of transport infrastructure(Molenaar 2005; Zayed et al. 2008; Kaliba et al. 2009; Doloi et al. 2012; Mahamid et al. 2012; Aziz and Abdel-Hakam 2016; Santoso and Soeng 2016;

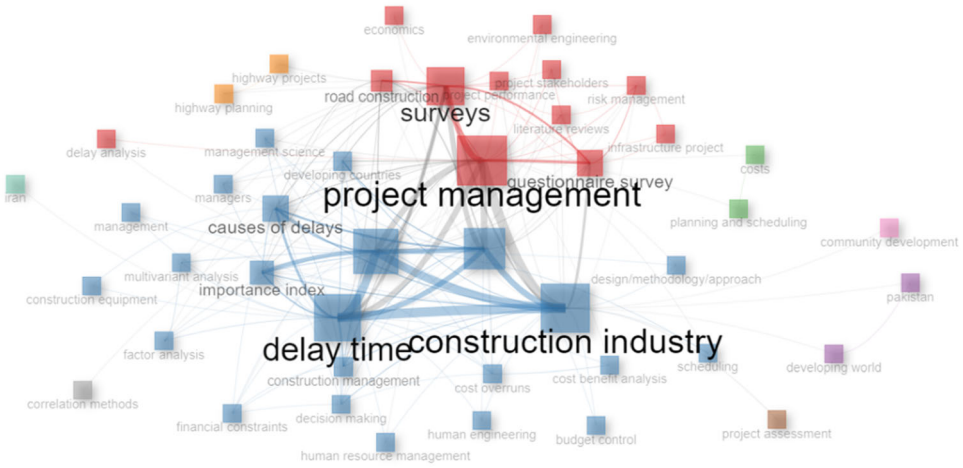


Figure 5. Word connection.

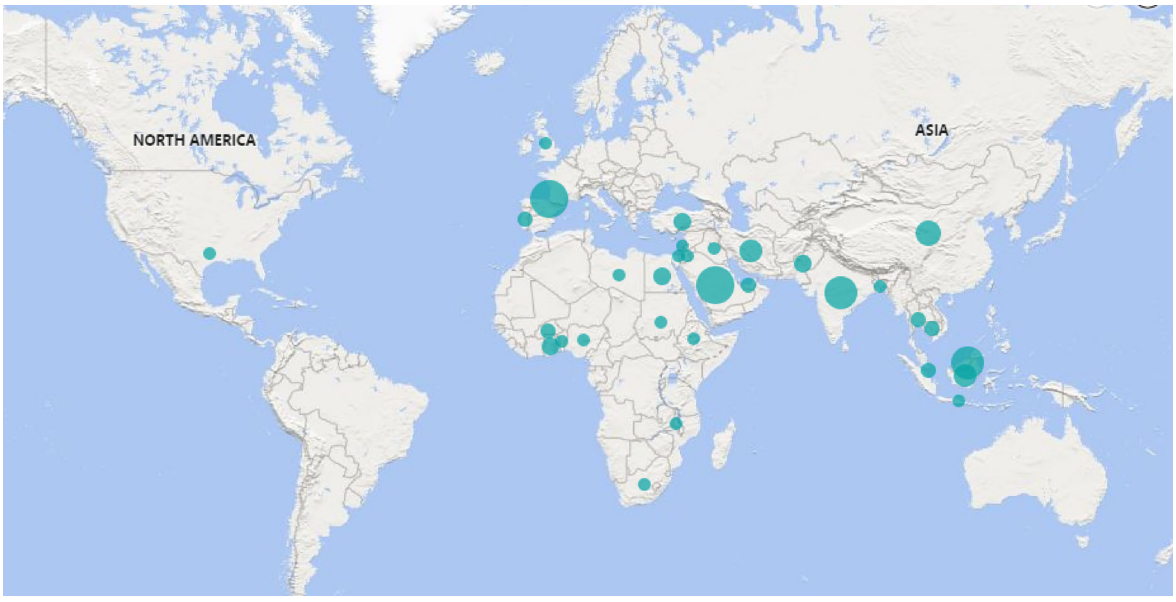


Figure 6. Distribution of case studies in articles.

Alfakhri et al. 2017; Amoatey and Ankrah 2017; Edwards et al. 2017; Hussain et al. 2017; Venkateswaran and Murugasan 2017; Al Hadithi 2018; Karunakaran et al. 2018; Khair et al. 2018; Thapanont et al. 2018; Antoniou 2021; Shukla et al. 2021; Abdurrahman et al. 2022; Karimi and Piroozfar 2022; Reddy and Nagakumar 2022; Yang et al. 2022; Abolelmagd et al. 2023; Ayudhya 2023; Setiani et al. 2023), other infrastructure (Yau and Yang 2012; Fallahnejad 2013; Wang and Yuan 2014; Hossen et al. 2015; Al-Hazim et al. 2017; Khan and Gul 2017; Wang and Yuan 2017; Abdullah et al. 2018; Kalebuka et al. 2018; Zarei et al. 2018; Al-Fadhali et al. 2019; Mbala et al. 2019; Sweis et al. 2019; Alenazi et al. 2022; Alshibani et al. 2022; Ismaila et al. 2022; Mahamid 2022; Sanjaya et al. 2022; Selvan et al. 2022; Hoque et al. 2023; Hussain et al. 2023; Kermanshachi and Pamidimukkala 2023; Khahro et al. 2023; Shash and AbuAlnaja 2023; Al-Gahtani et al. 2024), residential building (Asnaashari et al. 2009; Pourrostam and Ismail 2011; Doloi et al. 2012; Kraus 2012; Aziz 2013; Gündüz et al. 2013; Motaleb and Kishk 2013; Mydin et al. 2014; Sorooshian 2014; Arantes et al. 2015; Gunduz et al. 2015; Durdyev et al. 2017; Gebrehiwet and Luo 2017; Mpofu et al. 2017; Tarigan 2018; Wang et al. 2018; Wee et al. 2018; Chen et al. 2019; Abonassrya et al. 2022; Ajayi and Chinda 2022; Aravindhan et al. 2023; Irfanullah et al. 2023; Memon et al. 2023; Zadeh and Qayoumi 2023; Abdallah et al. 2024), and public building (Enshassi et al. 2009; Akogbe et al. 2013; Tawil et al. 2013; Arantes and Ferreira 2015; Shanmuganathan and Baskar 2015; Bagaya and Song 2016; Kim et al. 2016; Hussain et al. 2017; Sayed

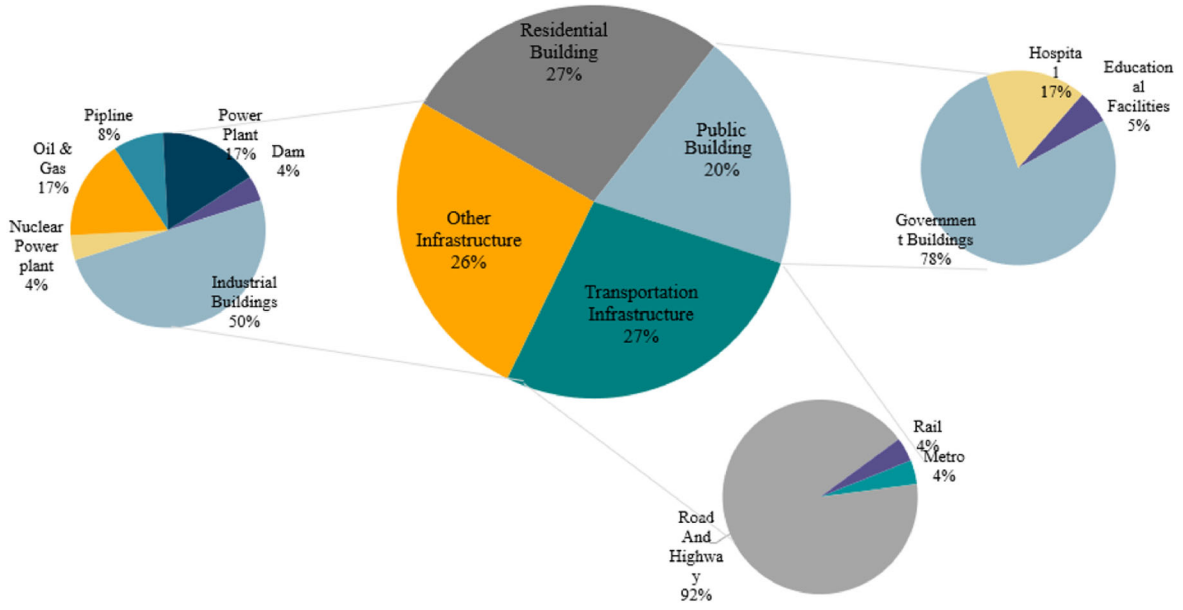


Figure 7. Frequency of articles on project delays by project type.

et al., 2017; 2017; Ametepey et al. 2018; Nawi et al. 2018; Zidane and Andersen 2018; Budayan 2019; Shivambu and Thwala 2019; Shrivas and Singla 2022; Demissew and Abiy, 2023; Ismail et al. 2023), are compared and grouped. The most common factors in different countries are presented.

Figure 7 illustrates the distribution of 92 selected articles examining the causes of delays in various construction projects. The analysis highlights the sectors with the most significant articles focus, with the hospital sector receiving the highest attention at 31%, indicative of its critical impact on public welfare. Conversely, sectors such as dams, nuclear power plants, and metro have received minimal attention at 1%, suggesting potential gaps in the literature. The graph serves to underscore the prioritization of research efforts in different construction sectors, providing a foundation for addressing delays more effectively.

The project type is categorized into 4 groups: private residential building, governmental building, transport infrastructure, and other infrastructure projects. The used model and project types are summarized in Figure 8.

Each article prioritizes the factors causing delay with different models and leads to different results. Hasanzadeh et al. (2021) compared 34 effective factors on the project time and based on their results, it showed that project management, time management, financial problems with contractors, and critical activities delay are the most important parameters on the project time (Hasanzadeh et al. 2011). In the study in New Zealand (2021), focusing on causes of delay in building projects, 14 types of delays were investigated with contractors. The project's inadequate planning and scheduling, ineffective site management and control, the contractor's poor communication skills, problems with resource management, the lack of necessary equipment for the job, and the contractor's poor coordination with technical staff were the most important of these causes (Babaeian Jelodar et al. 2021).

Figure 9 illustrates the distribution of different construction project types over several years. Each chart layer represents a project type, with the thickness indicating the number of projects in a given year. This visualization allows for easy identification of periods with high activity across multiple project types, offering a clear overview of the construction project landscape over time.

2.1.3. List of delay caused factors in literature

Based on the result of each article, the project delay can be affected by several factors. Based on the surveyed articles, all 104 factors extracted from the literature are categorized into 4 main groups Stakeholders, Resources, Processes, and External. In the next level, each group is divided into several

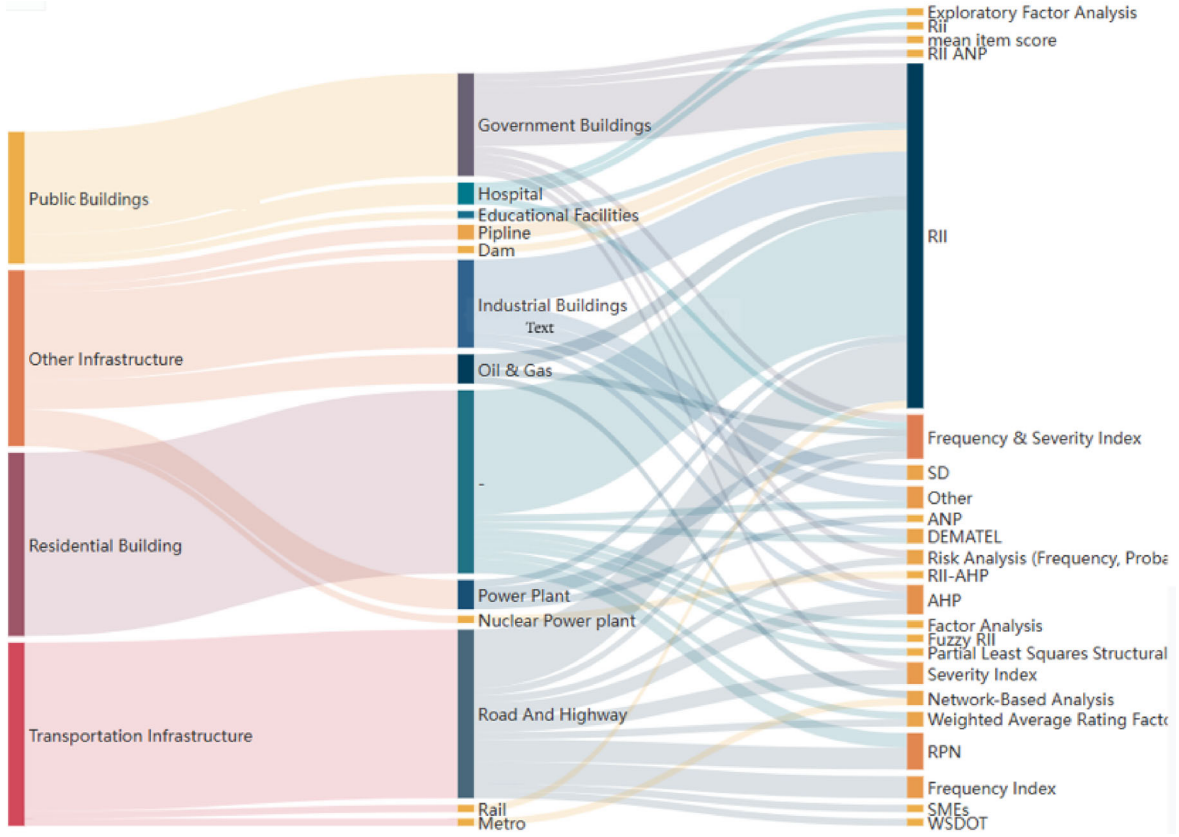


Figure 8. The most used models and project type.

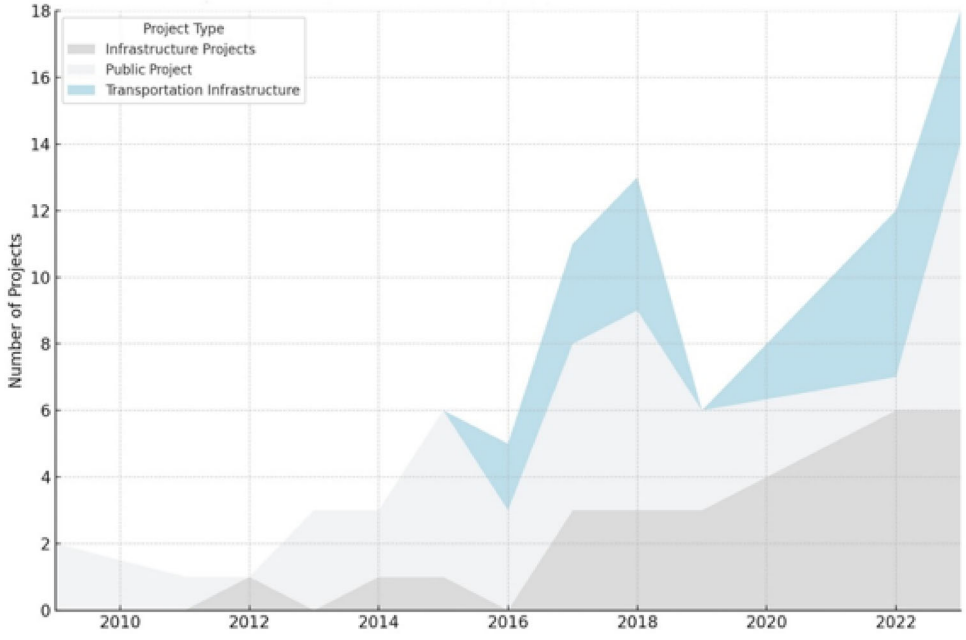


Figure 9. The number of projects and project type.

subgroups including Consultant (CN), Contractor (CT), Owner/Client (OW), Sub-contractor (SU), External Parties (EP), Equipment (EQ), Labour (LA), Material & Utility (MT), Engineering (EN), Design (DE), Tender and contract (TC), Communication (CO), Safety & Quality (SQ), Managerial (MA), External (EX). Since the grouping of articles varied in some cases, the most commonly used

grouping logic was applied to each factor. Table A-2 in the supplementary material presents a list of all factors presented in the selected papers and their ID which will be used in the remainder of this paper.

2.2. Phase 2: Data analysis (combining the results)

In this section, the results from the articles are combined to identify the highest-ranking factors for each country and project type. So, in the Excel Data base, for factors with the same meaning but different names, the most frequently used name across the selected studies is chosen as the standardized factor name. A new column is then added to this worksheet to include these unique factor names. Additionally, columns for factor codes and group codes are incorporated into the table to ensure consistency and facilitate further analysis. To connect these two tables and perform advanced analysis, a Pivot Table is created in Excel, and for the cleaning process, Power Query is adopted.

Various factors have been highlighted in numerous studies, but their relevance can vary depending on the type of project and geographic region. Our methodology aims to synthesize this data, providing a comprehensive analysis of the most critical factors impacting project timelines. In this study, the factors cited across different studies are aggregated. Given the vast array of literature, a filter to focus on factors that have been identified as significant in at least three separate articles is applied. This ensures that the analysis only includes factors with a demonstrated impact on project delays across multiple studies.

Since different studies used varying numbers of factors, the results of the articles were combined using a normalization process. First, the rank of each factor in each article was divided by the total number of factors considered in that article. Next, the normalized values of each factor across all articles were compared, and outliers (values beyond three standard deviations from the mean) were removed. Finally, the average of the remaining values was calculated, and the factors were ranked by sorting these averages in ascending order.the dataset is refined by considering the rank attributed to each factor within the studies. The rank is indicative of the factor's relative importance in causing delays, with a lower number representing a higher priority.

Each article in the dataset provides a vector of ranks for different factors. These vectors represent the rank of each factor within a specific study. The rank determined by the ith paper for the jth factors presented by kij, which kij can be any number between 1 to n (number of factors). Accordingly, the rank values of factors in the ith article can be described by xi as $X_i = [k_{i1}, ki_2, k_{i3} \ldots \ldots .k_{in}]$.

It is crucial to acknowledge that the importance of a specific factor may vary greatly depending on the study's context. For instance, political unrest might significantly influence project delays in one region but may be irrelevant in another. Our methodology accounts for this by identifying and removing such noise factors from the dataset.

For each factor, in each article, the ranks are normalized (let's call it as n_{ij}) by dividing its rank by the maximum rank of that article based on Equation (1).

$$n_{ij} = \frac{k_{ij}}{\theta} \forall i, j | \theta = \max\left(k_{ij}\right) \forall j \tag{1}$$

Then the average of normalized rank across all studies is calculated, using Equation (2). This average serves as a baseline for comparing the relative importance of each factor across different studies.

$$\overline{x_j} = \frac{\sum_{i=1}^{n} n_{ij}}{number\ \ of\ \ n_{ij} > 0} \forall j \tag{2}$$

The variance in normalized ranks for each factor to measure the consistency of its impact on project delays is also calculated. Factors with high variance may indicate noise, as their ranking significantly fluctuates between studies. These factors are candidates for removal to ensure the robustness of our analysis.

$$\frac{S_j^2 = \sum_{i=1}^{n} (n_{ij} - \overline{x_j})^2}{number\ \ of\ \ n_{ij} > 0 - 1 \forall j} \tag{3}$$

To determine a normal range for factor ranks, the average rank and variance are used to define upper and lower bounds. Any factor whose rank falls outside this range in any study is considered an outlier and removed from further analysis.

$$\left[\overline{x_j} - 3 * s_J, \overline{x_j} + *s_J\right] \tag{4}$$

After filtering out the noise, the median average of normalized rank for each factor from the remaining data is determined.

Finally, a rank index for each factor is developed by sorting them in descending order based on their median ranks. This rank index prioritizes factors according to their overall importance as indicated by multiple studies, providing a clear hierarchy of issues to be addressed in project delay mitigation efforts.

To clarify the process, the following numerical example is presented:

Consider two articles ranking the causes of delays for a specific project type in a given country. The first article identifies three factors, a, b, and c, with ranks of $a = 1$, $b = 2$, and $c = 3$. The second article includes four factors, a, b, c, and d, with ranks of $a = 1$, $b = 3$, $c = 4$, and $d = 2$.

To combine the results from these studies, the ranks must first be normalized. For the first study, the ranks are divided by the number of factors (3), and for the second study, the ranks are divided by the number of factors (4). The normalized ranks for each factor are as follows:

Factor a: (1/3, 1/4)
Factor b: (2/3, 3/4)
Factor c: (3/3, 4/4)
Factor d: (—, 2/4) (no value in the first study).
Next, the average and standard deviation of the normalized values are calculated for each factor:
Factor a: Average = 0.29, Standard Deviation = 0.04
Factor b: Average = 0.71, Standard Deviation = 0.04
Factor c: Average = 1.00, Standard Deviation = 0
Factor d: Average = 0.50, Standard Deviation =—(as there is only one value, no variation exists, and standard deviation is not needed).

Since all values fall within the normalized range, there is no noise, and the average values are used for ranking. Sorting the factors based on the average normalized rank results in the following order: a, d, b, c. Consequently, the final ranks are: $a = 1$, $d = 2$, $b = 3$, $c = 4$.

3. Result and discussion

In this study, 92 articles were selected to compare their results in ranking the causes of project delays across different countries.In this section, we first evaluate the critical risk factors across different project types within each country. This step involves identifying the primary delay causes unique to each country and project type. Next, the results from various countries are integrated to provide a comprehensive analysis of the delay causes across all project types. Finally, the results for all project types are combined, offering a holistic view of the global factors contributing to project delays.

3.1. Critical risk factors in different project types within each country

In this section, the results of previous studies in determining the highly important factors, causing project delay based on the project type are presented separately for each country. For this analysis, Microsoft Excel with two worksheets is used as the database. The first worksheet contains the selected articles, where each article is assigned a unique code. This code is listed in a row along with associated features such as author name, publication year, project type, studied countries, and methodology. The second worksheet contains a detailed list of factors used in each article. It includes columns for the article code, factor group, factor name, and factor rank.

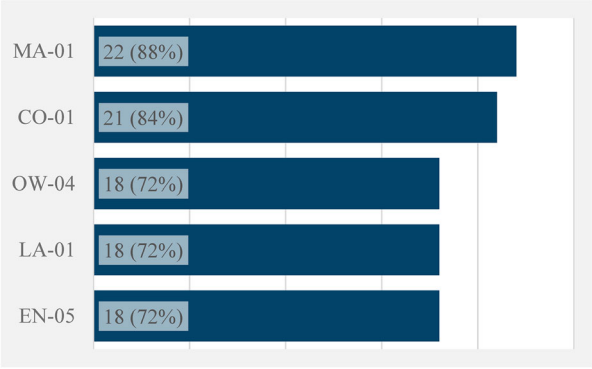


Figure 10. The most cited factors in articles related to residential building.

3.1.1. Residential building

In the research evaluating delays in residential building, a variety of factors contributing to project delays were analysed in different countries as following:

Cambodia: In Cambodia, the most significant factor causing project delays was identified as the shortage of materials on site. This was followed by unrealistic project scheduling and late delivery of materials. This suggests a critical need for better supply chain management and realistic scheduling in Cambodian projects (Durdyev et al. 2017).

China: Chinese projects highlight different primary concerns. While one study (Chen et al. 2019) places the shortage of adequate equipment and poor communication among contracting parties at the top, another (Wang et al. 2018) emphasizes delay in progress payments and variations in project scope as primary issues. This indicates a variance in project management challenges across different regions in China (Chen et al. 2019).

Egypt: The leading cause of project delays in Egypt was delay in progress payments, indicating funding problems. This was closely followed by different tactics or patterns for bribes and a shortage of equipment. These factors suggest significant financial and ethical challenges in the Egyptian construction industry (Aziz 2013).

India: In India, delays were most commonly caused by late material delivery by vendors, non-availability of drawings/designs on time, and financial constraints of contractors, highlighting issues in supply chain and financial management (Doloi et al. 2012).

Indonesia: Indonesian projects faced delays mainly due to requests for work changes during construction, delays in decision-making, and correction work due to contractor errors. This points to the need for better upfront planning and decision-making processes (Mpofu et al. 2017).

Iran: Iranian projects, across three different studies, consistently faced delays due to poor site management, delays in progress payments, and financial difficulties. These factors underscore the critical role of effective site management and stable financial planning in Iran (Asnaashari et al. 2009; Pourrostam and Ismail 2011).

Malaysia: In Malaysia, the primary causes of delay were conflicts between contractors and other parties, slow decision-making, and weather conditions. This indicates the importance of effective communication and adaptability to environmental conditions in Malaysian construction projects (Mydin et al. 2014; Sorooshian 2014; Wee et al. 2018).

Pakistan: In Pakistan, lack of coordination and poor contract management topped the list of delay causes, followed by delays in material procurement and shortage of funds. This points to the need for better organizational and financial management in Pakistani projects (Irfanullah et al. 2023).

Based on the articles, Figure 10 illustrates the most commonly cited factors across the surveyed articles, while Figure 11 highlights six factors that were ranked below third place in most articles for the residential building projects.

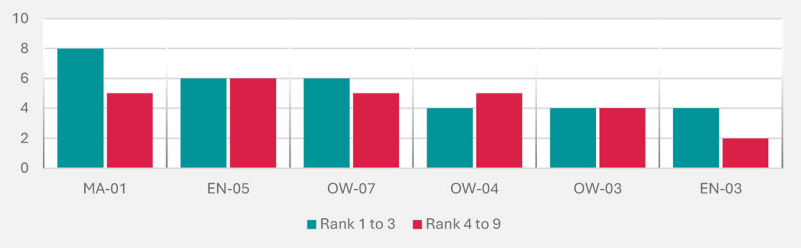


Figure 11. Factors ranked below third place in most articles related to residential building.

3.1.2. Public building

Based on the literature, a comparative analysis of the top factors contributing to public building project delays in various countries are presented below.

Benin: In Benin, the predominant factor causing project delays was financial capability (C11), consistently ranked at the top. This was followed by financial difficulties (C1) and poor subcontractor performance (C12). These findings suggest that financial stability and efficient subcontractor management are critical to project success in Benin (Akogbe et al. 2013).

Burkina Faso: The primary issues in Burkina Faso were the financial capability of the contractor and the financial difficulties of the owner, equally ranked as the top factors. This indicates that financial issues on both the contractor and owner sides are significant contributors to delays. Equipment availability was also a key factor, pointing to resource management issues (Bagaya and Song 2016).

Egypt: In Egypt, the leading cause of delay was finance and payments of completed work. This was followed by inadequate contractor experience and improper contractor planning, highlighting financial management and expertise as the main challenges (Sayed et al., 2017).

Ethiopia: The top-ranked factors in Ethiopia were inaccurate time estimate, force majeure, and the lowest bid procurement policy. These factors suggest challenges in accurate project estimation and external uncontrollable events impacting project timelines (Demissew and Abiy, 2023).

Gaza Strip: The most significant factors were strikes, military action, and border closures, followed by a lack of materials in markets and cash problems during construction. These findings indicate that external political and economic factors are major contributors to project delays in the Gaza Strip (Enshassi et al. 2009).

Ghana: In Ghana, delay in progress payments and variation orders during construction were the primary issues. Additionally, difficulties in financing projects and inflation were significant factors, emphasizing the importance of financial planning and stability (Amoatey et al. 2015; Ametepey et al. 2018).

India: Poor site management and supervision, ineffective project planning and scheduling, and delays in the arrival of materials were key issues in India. These factors underscore the need for effective site and project management (Shanmuganathan and Baskar 2015).

Malaysia: Project scope and design changes, poor planning, and bureaucracy in government agencies were the top factors in Malaysia, indicating challenges in project scope management and governmental processes (Nawi et al. 2018).

Pakistan: The primary factors in Pakistan were problems faced while travelling, increases in the price of consumer goods, and high transportation costs, reflecting the impact of external economic factors on project timelines (Hussain et al. 2017).

Portugal: Slow decision making by the developer, change orders, and unrealistic time schedules were the leading causes of delay in Portugal, highlighting challenges in decision-making and project planning (Arantes and Ferreira 2015).

South Africa: The main issues were problems with subcontractors, payment delays, and mistakes during the construction stage, emphasizing the need for efficient subcontractor management and financial stability (Shivambu and Thwala 2019).

Turkey: Law and regulation changes, improper evaluation and selection procedures, and urban plan changes were frequently cited, indicating that regulatory and planning issues are significant factors in Turkey (Budayan 2019).

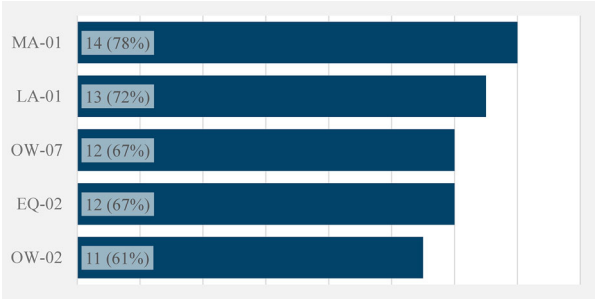


Figure 12. The most cited factors in articles related to public building.

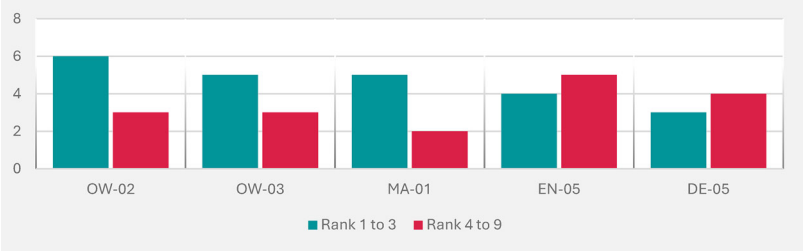


Figure 13. Factors ranked below third place in most articles related to public building.

United Kingdom: In the UK, insufficient capital, delays in getting progress payments, and contractor management problems were the top-ranked factors, highlighting the importance of financial management and contractor efficiency (Tawil et al. 2013).

Vietnam: Financial difficulties to the owner, insufficient data collection, lack of supervisor's responsibilities, and incompetence of contractors were the major factors in Vietnam, pointing towards the need for better financial management and improved project supervision (Kim et al. 2016).

Based on the articles, Figure 12 illustrates the most commonly cited factors across the surveyed articles, while Figure 13 highlights five factors that were ranked below third place in most articles for the public building projects.

3.1.3. Transport infrastructure

From the data provided in the literature, below is an analysis of the top factors contributing to project delays in various countries related to transport infrastructure projects.

Cambodia: The primary factor for delays in Cambodia is working during the rainy season, followed by flooding and the impact on people's land along the road construction project. These factors highlight the environmental and social issues affecting construction projects in Cambodia (Santoso and Soeng 2016).

China: In China, the major issues are problems with land expropriation, safety accidents, and slow decisions by clients, indicating that legal, safety, and management inefficiencies are significant contributors to delays (Yang et al. 2022).

Egypt: Owner financial problems and necessary variations rank highest in causing delays, with equipment shortages and previous working relationships of the owner also being notable. Contractor experience and unfair subcontractor relationships are also factors, indicating a mix of financial, management, and ethical issues (Aziz and Abdel-Hakam 2016).

Ghana: Delays in payment by owners, particularly the government, and changes in the scope by the owner during construction are the most cited issues. Inadequate contractor experience and unrealistic contract durations are also significant, showing a trend of governmental and contractor-related challenges (Amoatey and Ankrah 2017).

Greece: Adverse weather, late land acquisition, utility conflicts, and traffic control issues are the main causes of delays in Greek road projects (Antoniou 2021).

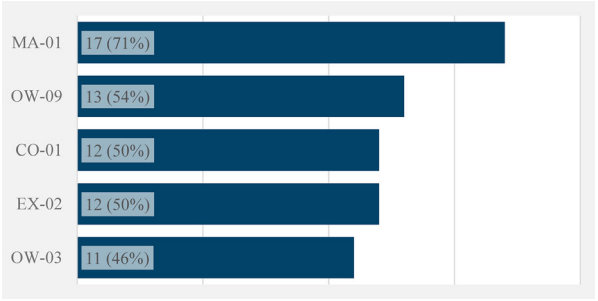


Figure 14. The most cited factors in articles related to transport infrastructure.

India: The slow deployment of labour on-site, delays due to the land acquisition process, and price variations are the top factors. Additionally, changes in the scope of contracts, claims and disputes, and community incursions highlight social and legal challenges (Venkateswaran and Murugasan 2017; Shukla et al. 2021).

Indonesia: Less quality control, late material delivery, and lack of coordination among project staff are the primary causes of delay, which points to management and operational inefficiencies (Abdurrahman et al. 2022).

Iraq: Political decisions and the country's economic crisis are the leading causes of delays, with material testing delays also contributing, emphasizing the impact of national stability and logistical processes (Al Hadithi 2018).

Libya: The main issues are delays in the conversion and transfer of utility services, difficulty in budget availability, and original contract durations being too short, highlighting financial and planning challenges (Alfakhri et al. 2017).

Malawi: Political situations and segmentation of the West Bank, causing limited movement between areas are the top factors, along with awarding projects to the lowest bid price, and delays in progress payments by the owner, indicating political and financial issues (Mahamid et al. 2012).

Malaysia: Improper planning, weather, and poor site management are cited as the main factors for project delays, suggesting the need for better strategic and operational management (Karunakaran et al. 2018).

Saudi Arabia: The weakness of financial and technical capabilities of some contractors, failure to study the site and tender properly, and problems of expropriation and removal of obstacles are the main issues, indicating technical and legal challenges (Abolelmagd et al. 2023).

Sudan: Problems with cash flow, delays in payment to the contractor, and difficulties in project financing are the most significant factors, reflecting severe financial management issues (Khair et al. 2018).

Thailand: Delays in progress payments, incomplete drawings, the financial status of the owner, lack of equipment efficiency, and delays in relocating existing utilities are the top issues, which points to financial and operational inefficiencies (Alfakhri et al. 2017; Thapanont et al. 2018).

Zambia: Delayed payment was identified as the leading cause of schedule delays in Zambia's road construction industry, with a 75% weighted occurrence. Other major causes included slow financial processes (67%), delayed fund release (60%), and contract modifications (57%). Additional issues like economic hardships, material procurement, staffing problems, and labour disputes ranged between 50%–56% (Kaliba et al. 2009).

Figure 14 illustrates the most commonly cited factors across the surveyed articles, while Figure 15 highlights five factors that were ranked below third place in most articles for the transport infrastructure projects.

3.1.4. Other infrastructures

Here is a synthesized comparison of factors contributing to other infrastructure project delays across various countries as indicated by the literature.

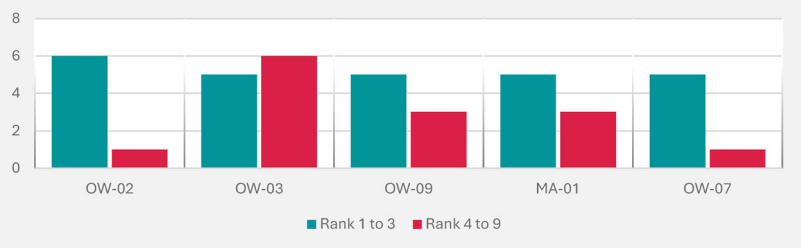


Figure 15. Factors ranked below third place in most articles related to transport infrastructure.

Bangladesh: Delay in progress payments was the leading cause of project delays, followed by rework due to mistakes during construction and delays in decision-making. This highlights financial and operational inefficiencies (Hoque et al. 2023).

Burkina Faso: The financial capability of the contractor and financial difficulties of the owner were both ranked as the most significant factors, indicating that financial stability is a major concern. Inadequate planning and slow payments were also notable, along with equipment availability (Khan and Gul 2017).

China: Construction changes initiated by clients, late monthly payments, and unreasonable schedule plans were the top-ranked factors, pointing to client management and financial scheduling as key issues (Wang and Yuan 2014).

India: The lockdown due to the pandemic was the most significant factor, followed by delays in progress payments and land acquisition issues. The pandemic's impact has been profound, affecting various aspects of construction projects (Selvan et al. 2022).

Iran: The low ability of contractors to provide imported material and slow land expropriation due to resistance from occupants were the most significant issues, followed by slow delivery of material by the client. This underscores supply chain and legal challenges in Iran (Fallahnejad 2013).

Jordan: Terrain conditions, weather conditions, and variation orders were ranked as the top factors affecting project timelines, highlighting environmental and change management challenges (Al-Hazim et al. 2017).

Malaysia: Delays in subcontractor's work, lack of subcontractor skill, and poor planning and scheduling were the primary concerns, pointing to the importance of skilled labour and project planning (Abdullah et al. 2018).

Nigeria: Corruption and bureaucracy in government were the leading causes of delays, followed by inadequate funding and price fluctuation/inflation, suggesting systemic and economic challenges (Ismaila et al. 2022).

Palestine: Poor labour productivity and coordination between construction parties were the most cited factors, with a lack of adequate manpower also contributing, indicating workforce management issues (Mahamid 2022).

Saudi Arabia: Lack of coordination/communication, ineffective planning and scheduling, client-related material issues, change order delays, and focus on financial analysis were the top factors. This suggests that communication, planning, and financial management are key areas of concern.

South Korea: Policy changes, uncompromising regulatory criteria, and robust design document review procedures were identified as the top factors, indicating regulatory and policy challenges (Hossen et al. 2015).

Zambia: Inadequate contractor experience and difficulties in financing the project by the client were the most significant factors, with the effect of rain on construction activities also highlighted, pointing to the need for skilled contractors and financial stability (Kalebuka et al. 2018).

Figure 16 illustrates the most commonly cited factors across the surveyed articles, while Figure 17 highlights six factors that were ranked below third place in most articles for other infrastructure projects.

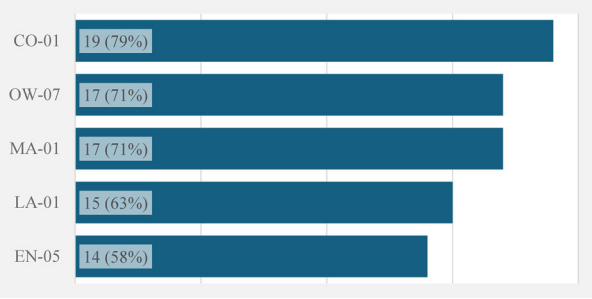


Figure 16. The most cited factors in articles related to other infrastructures.

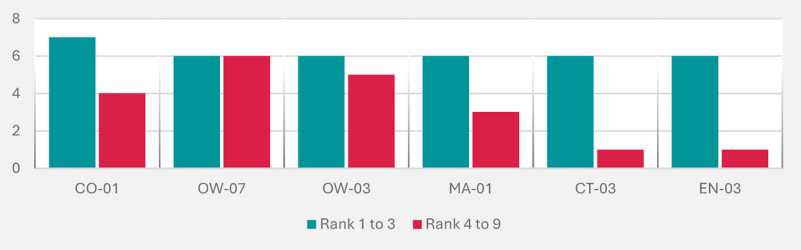


Figure 17. Factors ranked below third place in most articles related to other infrastructures.



Figure 18. Top 5 delay causes in countries for residential building.

3.2. Integration of country-specific results

Based on the ranking results of delay causes across different countries, this section delves into the integration of country-specific findings to identify the primary causes of delays in various project types. By synthesizing the results from multiple countries, we aim to highlight common trends and unique factors that contribute to project delays in different contexts. This integration allows us to extract key insights into the primary delay causes, offering a comparative analysis of how these factors vary across project types and geographical regions.

The factors contributing to delays across different project types are combined using the method explained in Section 2.2. The results are presented in Figures 18–21, which illustrate the top five delay factors identified for each project type in each country. These values are derived from the rankings provided in the surveyed articles, as outlined in the data analysis section. For a more detailed breakdown of each chart, please refer to Tables A-3 to A-6 in the supplementary material. These figures offer a visual comparison of the delay factors, providing a clearer understanding of how different causes of delays are ranked and how they vary across various project types.



Figure 19. Top 5 delay causes in countries for public building.

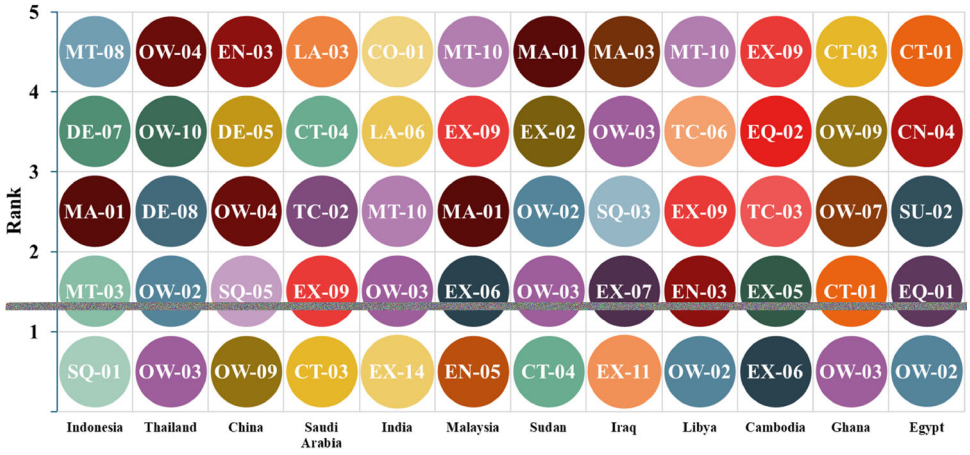


Figure 20. Top 5 delay causes in countries for transportation infrastructure.



Figure 21. Top 5 delay causes in the countries for other infrastructure.

3.2.1. Residential building

The analysis reveals that owner-related factors are the most frequently cited category among the top five delay causes across all countries for residential building. Design-related and contractor-related factors rank as the second and third most common groups, respectively. Together, these three categories account for 42% of the top five delay causes in residential building projects. The five most common delay causes for this project type are:

1. Owner requirement changes
2. Delays in contractor's payment by the owner
3. Contractor weakness in equipment procurement
4. Lack of coordination
5. Mistakes during the execution of works

On the other hand, consultant-related factors are never ranked among the top five causes of delays for this type of project in any country. External party-related factors are mentioned as a top-five cause only in Nigeria.

Among all the evaluated countries, the UAE stands out with a distinct ranking of delay factors. Of the top five ranked factors in the UAE, only one is common with other countries. The remaining four—delay in submitting documents by the contractor, shortage of labour, unclear and inadequate details in drawings, and lack of capability of the owner's representative—are unique to the UAE's residential building projects and appear only in its top five factors.

In Malaysia, weather conditions and contract-related factors are the main causes of delay. Additionally, two factors—rainfall and weather conditions, and poor site conditions with changes in topography—are included in the top five delay causes for Malaysia, but these factors are less significant for residential building in other countries.

3.2.2. Public building

For public building projects, owner-related, external, and contractor-related factors dominate the top five delay causes reported across countries, representing 56% of the total. The six most common delay causes in this category are:

1. Owner's financial problems
2. Delays in contractor's payment by the owner
3. Poor project time estimation
4. Changes in design
5. Contractor's financial problems
6. Poor and improper planning

On the other hand, communication- and consultant-related factors are never listed among the top five causes of delays in any country for this type of project (Figure 19).

3.2.3. Transportation infrastructure

In transportation infrastructure, the most frequent delay causes are also linked to owner-related, external, and contractor-related factors, accounting for 52% of the top five causes across countries. The six most commonly reported delay causes for this type of project are:

1. Delays in contractor's payment by the owner
2. Owner's financial problems
3. Poor site conditions and changes in topography
4. Poor project management and supervision
5. Owner's late decisions
6. Unavailability or delay in providing utilities on-site

On the other hand, consultant-, external party-, subcontractor-, and communication-related factors are rarely mentioned as major causes of delays in transportation infrastructures across the countries (Figure 20).

Table 2. Overlap of critical delay factors across all four categories.

Factors	Residential building	Public building	Transportation infrastructure	Other infrastructure
Delays in contractor's payment by the owner	✓	✓	✓	✓
Owner's financial problems		✓	✓	✓
Contractor's financial problems		✓		✓
Owner requirement changes	✓			✓
Poor project time estimation		✓		✓
Changes in design		✓		
Contractor weakness in equipment procurement	✓			
Lack of coordination	✓			
Mistakes during the execution of works	✓			
Poor and improper planning		✓		
Poor project management and supervision			✓	
Poor site conditions and changes in topography			✓	
Selection of inappropriate contractors/lowest bid procurement policy				✓
Shortage of labour				✓
Unavailability or delay in providing utilities on-site			✓	

3.2.4. Other infrastructure

For other infrastructure projects, owner-related, external, and contractor-related factors remain the most common among the top five delay causes, comprising 48% of the reported causes. The seven most frequently mentioned delay causes for this project type are:

1. Owner requirement changes
2. Delays in contractor's payment by the owner
3. Contractor's financial problems
4. Poor project time estimation
5. Shortage of labour
6. Owner's financial problems
7. Selection of inappropriate contractors/lowest bid procurement policy

On the other hand, management-, external party-, and consultant-related factors are rarely mentioned as major causes of delays in transportation infrastructures across the countries.

Table 2 summarizes the overlap of critical delay factors across all four categories. The comparison of critical delay factors across all four project categories reveals both common and unique patterns. Delays in contractor payments by the owner are a factor consistently present in every project type. Owner's and contractor's financial problems, along with owner requirement changes and poor project time estimation, also appear across multiple categories, highlighting financial and managerial challenges as pervasive issues. In contrast, certain factors are project-specific: residential projects face coordination problems, execution mistakes, and equipment procurement weaknesses; public buildings are more affected by design changes, poor planning, and inaccurate time estimation; transportation infrastructure suffers from poor site conditions, supervision challenges, and utility delays; while other infrastructure projects are uniquely impacted by lowest-bid procurement policies and labour shortages. This overlap underscores the central role of financial and owner-related issues, while also emphasizing the distinct challenges inherent to each project type.

3.3. Analysis of delay factors across all project types

Using the method presented in Section 2.2 to integrate the studies and extract the main delay causes across projects, regardless of their type or country, led to the identification of the top ten delay causes across all projects, which are: 1. Delays in contractor's payment by the owner, 2. Owner's financial problems, 3. Contractor's financial problems, 4. Poor project time estimation, 5. Owner requirement changes, 6. Change in design, 7. Poor site conditions and changes in topography, 8. Owner's late decisions, 9. Poor and improper planning, and 10. Selecting inappropriate contractors/lowest bid procurement policy.

Figure 22 illustrates the seven main factor groups and their frequency of occurrence among the top five delay causes across different countries. As shown, the owner-related factors group is the leading cause of delays in the projects.

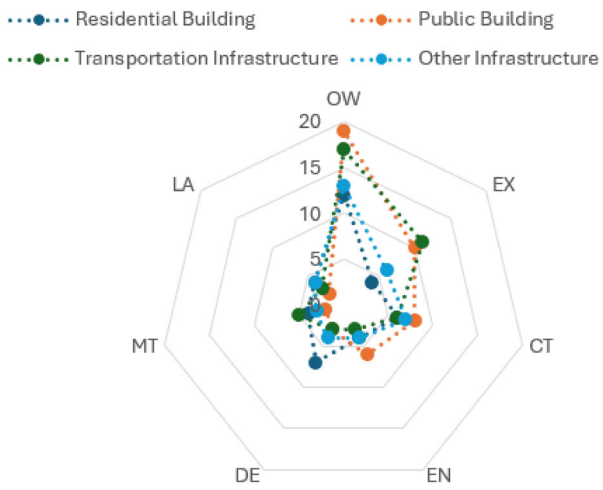


Figure 22. Main delay factors group.

3.4. Research limitation and future work

Like other research, this study also has some limitations. First, it focuses on the most recently published papers, neglecting changes in factor rankings over time. Additionally, countries are evaluated separately without being categorized based on their location or level of development. Another limitation is that global situations, such as the impact of the pandemic, are not considered. Moreover, while the projects are categorized into four groups, the general size of the projects is not taken into account. Finally, the area of rail and metro within transportation infrastructure projects receives less attention in this research.

To address these limitations, several areas for future work are recommended. A valuable direction would be to compare changes in the ranking of risk factors over time, particularly during specific global economic downturns or pandemics compared to peacetime. Another approach would be to categorize countries into groups and compare the ranking of risk factors in developed, developing, and underdeveloped countries. Additionally, identifying key delays in countries that have experienced war could provide further insights. Future studies should also examine the impact of project size on key delay causes and consider the relationship between delay causes and the maturity levels of contractors or owner organizations.

4. Conclusion

This study provides an in-depth analysis of the primary causes of delays in construction projects across 35 countries, focusing on four main project types: residential buildings, public projects, transportation infrastructure, and other infrastructure projects. Utilizing the PRISMA method, this research integrates findings from 92 selected articles to identify and rank the top delay factors in construction projects.

The primary delay causes were categorized into four key groups—Stakeholders, Resources, Processes, and External factors—each further subdivided into specific factors such as Consultant, Contractor, Owner/Client, Sub-contractor, Equipment, Labour, Material & Utility, Engineering, Design, Tender and Contract, Communication, Safety & Quality, Managerial, and External Parties.

While the causes of delays vary across different project types and countries, some factors are common across multiple regions, with certain delay causes recurring in several countries. Based on the studies, the top five delay causes in various projects across these countries are as follows:

For the residential building projects: 1: Owner requirement changes, 2: Delays in contractor's payment by the owner, 3: Contractor weakness in equipment procurement, 4: Lack of coordination, and 5: Mistakes during the execution of works.

For the public building projects: 1: Owner's financial problems, 2: Delays in contractor's payment by the owner, 3: Poor project time estimation, 4: Changes in design, and 5: Contractor's financial problems.

For the transportation infrastructure: 1: Delays in contractor's payment by the owner, 2: Owner's financial problems, 3: Poor site conditions and changes in topography, 4: Poor project management and supervision, and 5: Owner's late decisions

For other infrastructure: 1: Owner requirement changes, 2: Delays in contractor's payment by the owner, 3: Contractor's financial problems, 4: Poor project time estimation, and 5: Shortage of Labor.

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Data availability statement

The data that support the findings of this study are available on request from the corresponding author.

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